

一、Langchain-Chatchat 部署搭建 v0.2.10

参考页面:

<https://github.com/chatchat-space/Langchain-Chatchat/blob/v0.2.10/README.md>

<https://github.com/chatchat-space/Langchain-Chatchat/wiki/>

NOTE:

准备一个 GPU 的服务器，至少 24G+，建议直接选择阿里云的 DSW 的 ecs.gn7i-c8g1.2xlarge；如果选择 DSW 建议镜像选择: modelscope:1.14.0-pytorch2.1.2-gpu-py310-cu121-ubuntu22.04

—> 如果你只是使用一些小的模型，那么可以考虑使用 CPU 环境。

如果选择 CPU 环境，建议选择一个阿里云的 ECS 4 核 16G 的机器: ecs.e-c1m4.xlarge；

NOTE: 如果选择的环境没有安装 python 环境，需要安装 python 环境，建议直接安装 mini-conda。

▼ 安装 python

Bash

```
wget https://repo.anaconda.com/miniconda/Miniconda3-py310_24.9.2-0-Linux-x86_64.sh
```

```
sh Miniconda3-py310_24.9.2-0-Linux-x86_64.sh
```

安装完成后，退出重新进入

1. 从 github 官网下载对应的源码 [Langchain-Chatchat](#)

Bash

```
git clone https://github.com/chatchat-space/Langchain-Chatchat.git
```

或者

```
git clone https://gitee.com/shanghaileige/Langchain-Chatchat.git
```

```
cd Langchain-Chatchat
```

```
git tag -l
```

```
git fetch --tags
```

```
git checkout v0.2.10
```

2. 安装依赖环境，可以创建一个虚拟环境

Bash

PS: 内部的版本可以稍微修改一下，但是依赖比较麻烦，需要修改相关库的版本

pip install --use-pep517 -r requirements.txt 异常时候的执行命令

<http://mirrors.cloud.aliyuncs.com/pypi/simple/>

```
pip install -r requirements.txt
```

```
pip install -r requirements_api.txt
```

```

pip install -r requirements_webui.txt
# 在三个txt文件安装完成后，重新安装一下transformers ( QWEN模型要求的版本更高 )
pip install transformers==4.37.0

# huggingface-hub版本不要过高，最好为...
pip install huggingface-hub==0.20.3

```

3. 从网络下载模型文件(huggingface 或者 modelscope)

Python

```

# 将下载的模型移动到Langchain-Chatchat根目录(默认情况)，并且文件夹名称必须和config的配置对应上
# 或者下载到某个文件夹下，并下载config中进行给定修改

```

```

# 方式一：

```

```

# 从huggingface下载模型

```

```

import os
os.environ['HF_ENDPOINT'] = 'https://hf-mirror.com'
from huggingface_hub import snapshot_download
print(snapshot_download("moka-ai/m3e-base"))
print(snapshot_download("zai-org/chatglm2-6b"))
print(snapshot_download("zai-org/chatglm3-6b"))
print(snapshot_download("qwen/Qwen1.5-0.5B-Chat"))
### 下载完成后，从下载完成的路径将模型复制到你准备放模型的路径下

```

```

## 方式二：

```

```

# 从modelscope下载模型

```

```

from modelscope import snapshot_download
print(snapshot_download("thomas/m3e-base"))
print(snapshot_download("ZhipuAI/chatglm2-6b"))
print(snapshot_download("ZhipuAI/chatglm3-6b"))
print(snapshot_download("qwen/Qwen1.5-0.5B-Chat"))

# 移动下载到的模型到Langchain-Chatchat根目录
cp -R /mnt/workspace/.cache/modelscope/thomas/m3e-base .
cp -R /mnt/workspace/.cache/modelscope/ZhipuAI/chatglm2-6b .
cp -R /mnt/workspace/.cache/modelscope/ZhipuAI/chatglm3-6b .
cp -R /mnt/workspace/.cache/modelscope/qwen/Qwen1__5-0__5B-Chat .

```

```

# 方式三：

```

```

# 或者直接执行如下下载命令

```

```

apt-get update
apt-get install git-lfs
git lfs install

git clone https://www.modelscope.cn/thomas/m3e-base.git
git clone https://www.modelscope.cn/ZhipuAI/chatglm2-6b.git
git clone https://www.modelscope.cn/ZhipuAI/chatglm3-6b.git
git clone https://www.modelscope.cn/Qwen/Qwen1.5-0.5B-Chat.git
git clone https://www.modelscope.cn/Qwen/Qwen3.5-0.8B.git

```

```
git clone https://hf-mirror.com/moka-ai/m3e-base
git clone https://hf-mirror.com/Qwen/Qwen1.5-0.5B-Chat
git clone https://hf-mirror.com/zai-org/chatglm2-6b
git clone https://hf-mirror.com/zai-org/chatglm3-6b
```

PS: 可以考虑将文件夹中的点换成三个下划线

4. 创建基础配置文件，并修改基础配置信息

Bash

s# 执行下列命令，生成相关的config.py文件
python copy_config_example.py

修改信息如下

1. configs/basic_config.py文件内容:

24行修改为: `os.makedirs(LOG_PATH, exist_ok=True)`

2. configs/model_config.py文件内容:

PS: 如果模型不是放到根目录，那么修改修改参数: `MODEL_ROOT_PATH`

9行修改为: `EMBEDDING_MODEL = "m3e-base"`

29行修改为:

`LLM_MODELS = ["Qwen1___5-0___5B-Chat", "xinghuo-api", "deepseekv3-api"]`

180行增加:

`"Qwen1___5-0___5B-Chat": "Qwen/Qwen1.5-0.5B-Chat",`

`"Qwen3___5-0___8B": "Qwen/Qwen3.5-0.8B",`

3. configs/kb_config.py文件内容:

52行修改为: `SENIVERSE_API_KEY = "SqWCdI5TuUyD4Nbby"` # 需要先申请心知天气的API KEY

4. (可选) 针对在线模型xinghuo-api，进行如下操作:

参考页面: <https://www.xfyun.cn/doc/spark/Web.html>

4.1 在xinghuo模型页面进行个人认证、账号申请及资源购买

4.2 修改model_config.py文件内容:

#=====

```
"xinghuo-api": {
    "APPID": "xxx",
    "APISecret": "xxxx",
    "api_key": "xxxx",
    "version": "v4.0", # 你使用的讯飞星火大模型版本，可选包括 "v3.0", "v2.0", "v1.5"
    "provider": "XingHuoWorker",
},
```

#=====

4.2 修改server/model_workers/xinghuo.py/XingHuoWorker/do_chat方法里面的内容:

#=====

```
version_mapping = {
    "v1.5": {
        "domain": "general",
        "url": "ws://spark-api.xf-yun.com/v1.1/chat",
        "max_tokens": 4000
    },
    "v2.0": {
        "domain": "generalv2",
        "url": "ws://spark-api.xf-yun.com/v2.1/chat",
        "max_tokens": 8000
    },
}
```

```

    },
    "v3.0": {
        "domain": "generalv3",
        "url": "wss://spark-api.xf-yun.com/v3.1/chat",
        "max_tokens": 8000
    },
    "v4.0": {
        "domain": "Ultra",
        "url": "wss://spark-api.xf-yun.com/v4.0/chat",
        "max_tokens": 8000
    },
}
#=====

```

5. (可选)集成DeepSeek API,进行如下操作: --> 兼容支持所有支持OpenAI API接口规范的模型

参考页面: <https://api-docs.deepseek.com/zh-cn/>

5.1 在DeepSeek申请api_keys,以及充值;

5.2 修改model_config.py文件内容

5.2.1 在ONLINE_LLM_MODEL里面增加

```

# =====
"deepseekv3-api": {
    "model_name": "deepseek-chat",
    "api_base_url": "https://api.deepseek.com/v1",
    "api_key": "sk-xxxx",
    "openai_proxy": "",
    "openai-api": True
},
# =====

```

5.2.2 在LLM_MODELS里面增加 deepseekv3-api

5.3 修改server/utils.py/get_ChatOpenAI和server/utils.py/get_OpenAI方法内容:

```

# =====
if model_name == "openai-api" or config.get("openai-api", False):
# =====

```

5.4 修改configs/server_config.py中的端口号

```

# ===== FSCAT_MODEL_WORKERS
"deepseekv3-api": {
    "port": 21011,
},
# =====

```

6. (可选)为了可以通过docs页面查看接口,修改server/utils.py/MakeFastAPIOffline

直接将这个方法里面的内容注释掉即可,注释所有代码后,增加一个pass

5. (可选)构建初步知识 embedding 内容,构建索引

Bash

```
# 需要一定的资源的
python init_database.py --recreate-vs
```

6. 启动服务

```
Bash

python startup.py -a

# 启动日志如下

root@dsw-930292-596546d9cc-nn855:/mnt/workspace/llm03182/Langchain-Chatchat# python st
artup.py -a

=====Langchain-Chatchat Configuration=====
=====
操作系统 : Linux-5.10.134-17.2.1.lifsea8.x86_64-x86_64-with-glibc2.35.
python版本 : 3.10.13 (main, Sep 11 2023, 13:44:35) [GCC 11.2.0]
项目版本 : v0.2.10
langchain版本 : 0.0.354. fastchat版本 : 0.2.35

当前使用的分词器 : ChineseRecursiveTextSplitter
当前启动的LLM模型 : ['Qwen1___5-0___5B-Chat', 'xinghuo-api'] @ cuda
{'device': 'cuda',
 'host': '0.0.0.0',
 'infer_turbo': False,
 'model_path': 'Qwen1___5-0___5B-Chat',
 'model_path_exists': True,
 'port': 20002}
{'APISecret': '',
 'APPID': '',
 'api_key': '',
 'device': 'auto',
 'host': '0.0.0.0',
 'infer_turbo': False,
 'online_api': True,
 'port': 21003,
 'provider': 'XingHuoWorker',
 'version': 'v3.0',
 'worker_class': <class 'server.model_workers.xinghuo.XingHuoWorker'>}}
当前Embeddings模型 : m3e-base @ cuda
=====Langchain-Chatchat Configuration=====
=====

2025-03-18 20:45:57,316 - startup.py[line:655] - INFO: 正在启动服务 :
2025-03-18 20:45:57,316 - startup.py[line:656] - INFO: 如需查看 llm_api 日志, 请前往 /mn
t/workspace/llm03182/Langchain-Chatchat/logs
/opt/conda/lib/python3.10/site-packages/langchain_core/_api/deprecation.py:117: LangCh
ainDeprecationWarning: 模型启动功能将于 Langchain-Chatchat 0.3.x重写, 支持更多模式和加速启动 ,
```

0.2.x中相关功能将废弃

warn_deprecated(

```
2025-03-18 20:46:01 | INFO | model_worker | Register to controller
2025-03-18 20:46:01 | ERROR | stderr | INFO: Started server process [6609]
2025-03-18 20:46:01 | ERROR | stderr | INFO: Waiting for application startup.
2025-03-18 20:46:01 | ERROR | stderr | INFO: Application startup complete.
2025-03-18 20:46:01 | ERROR | stderr | INFO: Uvicorn running on http://0.0.0.0:20000 (Press CTRL+C to quit)
```

```
2025-03-18 20:46:02 | INFO | model_worker | Loading the model ['Qwen1___5-0___5B-Chat'] on worker 01919fd0 ...
```

Special tokens have been added in the vocabulary, make sure the associated word embeddings are fine-tuned or trained.

```
2025-03-18 20:46:12 | INFO | model_worker | Register to controller
INFO: Started server process [6754]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:7861 (Press CTRL+C to quit)
```

=====Langchain-Chatchat Configuration=====

操作系统 : Linux-5.10.134-17.2.1.lifsea8.x86_64-x86_64-with-glibc2.35.
python版本 : 3.10.13 (main, Sep 11 2023, 13:44:35) [GCC 11.2.0]
项目版本 : v0.2.10
langchain版本 : 0.0.354. fastchat版本 : 0.2.35

当前使用的分词器 : ChineseRecursiveTextSplitter

当前启动的LLM模型 : ['Qwen1___5-0___5B-Chat', 'xinghuo-api'] @ cuda

```
{'device': 'cuda',
 'host': '0.0.0.0',
 'infer_turbo': False,
 'model_path': 'Qwen1___5-0___5B-Chat',
 'model_path_exists': True,
 'port': 20002}
{'APISecret': '',
 'APPID': '',
 'api_key': '',
 'device': 'auto',
 'host': '0.0.0.0',
 'infer_turbo': False,
 'online_api': True,
 'port': 21003,
 'provider': 'XingHuoWorker',
 'version': 'v3.0',
 'worker_class': <class 'server.model_workers.xinghuo.XingHuoWorker'>}
```

当前Embeddings模型 : m3e-base @ cuda

服务端运行信息 :

OpenAI API Server: http://127.0.0.1:20000/v1
Chatchat API Server: http://127.0.0.1:7861

Chatchat WEBUI Server: <http://0.0.0.0:8501>

=====Langchain-Chatchat Configuration=====

Collecting usage statistics. To deactivate, `set browser.gatherUsageStats` to `False`.

You can now view your Streamlit app [in](#) your browser.

URL: <http://0.0.0.0:8501>

UI界面:

<http://8.136.193.178:8501/>

后端服务接口页面:

<http://8.136.193.178:7861/docs>

OpenAI接口页面:

<http://8.136.193.178:20000/docs>

Controller模型管理页面:

<http://8.136.193.178:20001/docs>

Work具体模型页面:

7. 内部主要依赖的库

Python

langchain --> 思维链的库
transformers --> 大语言模型恢复加载预测的基础库
uvicorn --> http/接口相关的库
fschat/fastapi/openai --> 部署/http/接口相关的库
streamlit --> web页面相关库
asyncio --> python异步处理基础库
 asynico是python的异步处理框架库
faiss-cpu --> 向量检索库

8. Langchain-Chatchat 主要构成部分:

Markdown

1. Web页面: 默认端口8501
前端展示聊天页面, 并将用户输入的问题发送给后端API, 并接受后端API的输出结果, 并显示到页面上。
2. 后端API: 默认端口7861
接收前端传递的参数
基于业务逻辑将参数进行整合处理, 封装成对象
调用OpenAI API接口, 获取LLM的返回结果
3. OpenAI API规范的接口: 默认端口20000

接收后端API服务传递过来的参数

通过Controller获取LLM Worker对应的服务地址

将参数透传给LLM Worker服务

4. Controller服务：默认端口20001

负责LLM Worker的注册、状态监听、Worker获取等内容

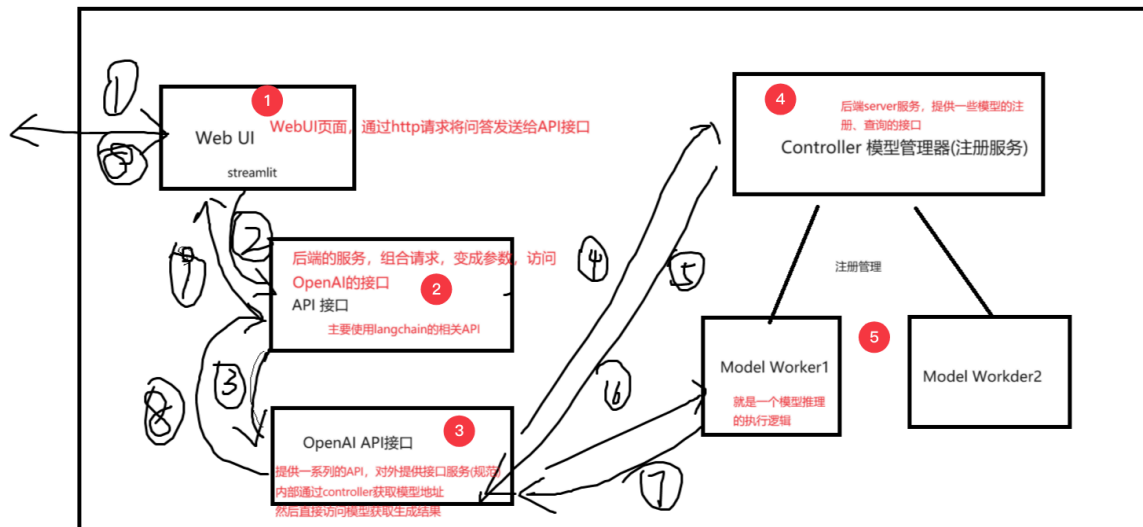
5. LLM Worker服务：默认端口20002

负责具体的LLM的调用 --> 和基于transformers库的模型恢复、预测代码逻辑基本一致

NOTE: 可以支持存在多个LLM Worker服务

NOTE: 每个模块与模块之间的通信/数据传递全部使用http请求来实现

LangChain-ChatChat的执行链路



异常说明:

1. pip 安装过程中提示_DeprecatedInstaller: setuptools.installer and fetch_build_eggs are deprecated.

解决方案：基于异常提示，在安装时候使用额外参数即可：pip install --use-pep517 -r requirements.txt

```
Collecting wavedrom (from markdown2[all]>fchat==0.2.35->-r requirements.txt (line 10))
  Downloading http://mirrors.cloud.aliyuncs.com/pypi/packages/be/71/6739e3abac630540aaeaece4584c39f8b5f8658ce6ca517efec455e3de/wavedrom-2.0.3.post3.tar.gz (137 kB)
  Preparing metadata (setup.py) ... error
error: subprocess-exited-with-error

python setup.py egg_info did not run successfully.
exit code: 1
[63 lines of output]
/root/.miniconda3/lib/python3.10/site-packages/setuptools/dist.py:452: SetuptoolsDeprecationWarning: Invalid dash-separated options
!!

*****
Usage of dash-separated 'index-url' will not be supported in future
versions. Please use the underscore name 'index_url' instead.

This deprecation is overdue, please update your project and remove deprecated
calls to avoid build errors in the future.

See https://setuptools.pypa.io/en/latest/userguide.html for details.
*****

!!
opt = self.warn_dash_deprecation(opt, section)
/root/.miniconda3/lib/python3.10/site-packages/setuptools/_init_.py:94: _DeprecatedInstaller: setuptools.installer and fetch_build_eggs are deprecated.
!!

*****
Requirements should be satisfied by a PEP 517 installer.
If you are using pip, you can try 'pip install --use-pep517'.
*****

!!
```

2. pip 安装库时候异常，提示版本依赖异常：pip's dependency

解决方案：暂时不考虑这种警告，先运行，如果后期报错，那么再考虑版本依赖的问题。

ERROR: pip's dependency resolver does **not** currently take into account **all** the packages that are installed. This behaviour **is** the source of the following dependency conflict s.

pai-easycv **0.11.6** requires timm==**0.5.4**, but you have timm **0.9.16** which **is** incompatible.

3. 模型下载异常

解决方案:

方案 1: 手动下载模型放到对应文件夹即可

方案 2: 修改 modelscope/hub/file_download.py 中的 http_get_file 方法中的长度判断的代码, 进行注释即可。

```
[3] print(snapshot_download("thomas/m3e-base"))

Downloading: 429KB [00:00, 38.1MB/s]
2024-12-13 11:06:24,841 - modelscope - ERROR - File tokenizer.json download incomplete, content_length: None but the file downloaded length: 439124, please download again

FileDownloadError: File tokenizer.json download incomplete, content_length: None but the file downloaded length: 439124, please download again

Hide Error Details
-----
FileDownloadError: Traceback (most recent call last)
Cell In[3], line 1
----> 1 print(snapshot_download("thomas/m3e-base"))

File ~/opt/conda/lib/python3.10/site-packages/modelscope/hub/snapshot_download.py:156, in snapshot_download(model_id, revision, cache_dir, user_agent, local_files_only, cookies, ignore_file_pattern)
    154         parallel_download(
    155             url,
    156             temp_cache_dir,
    157             cookies=cookies,
    158             file_size=model_file["size"])
    159     except:
    160         http_get_file(
    161             url,
    162             temp_cache_dir,
    163             model_file["name"],
    164             headers=headers,
    165             cookies=cookies)
    166     # check file integrity
    167     temp_file = os.path.join(temp_cache_dir, model_file["name"])

File ~/opt/conda/lib/python3.10/site-packages/modelscope/hub/file_download.py:345, in http_get_file(url, local_dir, file_name, cookies, headers)
    343     msg = 'File %s download incomplete, content_length: %s but the \
    344         file downloaded length: %s, please download again' % (
    345         file_name, total, downloaded_length)
    346     logger.error(msg)
--> 347     FileDownloadError(msg)
    348     os.replace(temp_file_name, os.path.join(local_dir, file_name))

FileDownloadError: File tokenizer.json download incomplete, content_length: None but the file downloaded length: 439124, please download again
```

```
except (Exception) as e: # no matter what happens, we will retry.
    retry = retry.increment('GET', url, error=e)
    retry.sleep()

logger.debug('Saving %s to cache at %s', url, local_dir)
downloaded_length = os.path.getsize(temp_file_name)
if total != downloaded_length:
    # os.remove(temp_file_name)
    msg = 'File %s download incomplete, content_length: %s but the \
    file downloaded length: %s, please download again' % (
    file_name, total, downloaded_length)
    logger.error(msg)
    raise FileDownloadError(msg)
    os.replace(temp_file_name, os.path.join(local_dir, file_name))

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```

4. Embedding 初始化时候异常:

原因: 模型错误, 模型没有下载成功。

```

root@autodl-container-adf4419373-balbf57: /Langchain-Chatchat# python init_database.py --recreate-vs
2025-01-10 15:10:23.554 - utils.py[line:145] - INFO: Note: detected 128 virtual cores but NumExpr set to maximum of 64, check "NUMEXPR_MAX_THREADS" environment variable.
2025-01-10 15:10:23.554 - utils.py[line:148] - INFO: Note: NumExpr detected 128 cores but "NUMEXPR_MAX_THREADS" not set, so enforcing safe limit of 8.
recreating all vector stores
2025-01-10 15:10:23.957 - faiss_cache.py[line:92] - INFO: loading vector store in 'samples/vector_store/m3e-base' from disk.
2025-01-10 15:10:24.511 - SentenceTransformer.py[line:86] - INFO: Load pretrained SentenceTransformer: m3e-base
2025-01-10 15:10:24.511 - SentenceTransformer.py[line:805] - WARNING: No sentence-transformers model found with name m3e-base. Creating a new one with MEAN pooling.
2025-01-10 15:10:25.126 - embeddings_api.py[line:39] - ERROR: Error while deserializing header: HeaderTooLarge
AttributeError: 'NoneType' object has no attribute 'conjugate'

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "/root/Langchain-Chatchat/init_database.py", line 107, in <module>
    folder2db(kb_names=args.kb_name, mode="recreate_vs", embed_model=args.embed_model)
  File "/root/Langchain-Chatchat/server/knowledge_base/migrate.py", line 121, in folder2db
    kb.create_kb()
  File "/root/Langchain-Chatchat/server/knowledge_base/kb_service/base.py", line 80, in create_kb
    self.do_create_kb()
  File "/root/Langchain-Chatchat/server/knowledge_base/kb_service/faiss_kb_service.py", line 51, in do_create_kb
    self.load_vector_store()
  File "/root/Langchain-Chatchat/server/knowledge_base/kb_service/faiss_kb_service.py", line 28, in load_vector_store
    return kb.faiss_pool.load_vector_store(kb_name=self.kb_name)
  File "/root/Langchain-Chatchat/server/knowledge_base/kb_cache/faiss_cache.py", line 102, in load_vector_store
    vector_store = self.new_vector_store(embed_model=embed_model, embed_device=embed_device)
  File "/root/Langchain-Chatchat/server/knowledge_base/kb_cache/faiss_cache.py", line 60, in new_vector_store
    vector_store = FAISS.from_documents([doc], embeddings, normalize_L2=True, distance_strategy="METRIC_INNER_PRODUCT")
  File "/root/miniconda3/lib/python3.10/site-packages/langchain_core/vectorstores.py", line 508, in from_documents
    return cls.from_texts(texts, embedding, metadatas=metadatas, **kwargs)
  File "/root/miniconda3/lib/python3.10/site-packages/langchain_community/vectorstores/faiss.py", line 965, in from_texts
    embeddings = embedding.embed_documents(texts)
  File "/root/Langchain-Chatchat/server/knowledge_base/kb_service/base.py", line 420, in embed_documents
    return normalize(embeddings).tolist()
  File "/root/Langchain-Chatchat/server/knowledge_base/kb_service/base.py", line 37, in normalize
    norm = np.linalg.norm(embeddings, axis=1)
  File "<array function internals>", line 200, in norm
  File "/root/miniconda3/lib/python3.10/site-packages/numpy/linalg/linalg.py", line 2541, in norm
    s = (x.conj() * x).real
TypeError: loop of ufunc does not support argument 0 of type NoneType which has no callable conjugate method

```

5. ml_dtypes 库异常

原因：在启动或者初始化知识库的过程中，报 cannot import name 'float4_e2m1fn' from 'ml_dtypes' 异常，并且导致执行失败的情况；

解决办法：安装一个更高版本的 ml_dtypes 库: pip install ml_dtypes==0.5.0

6. 进行知识库创建时候出现异常：

ImportError: libGL.so.1: cannot open shared object file: No such file or directory

原因：OpenCV 依赖的 linux 的底层文件没有安装好导致的异常信息

解决办法：ubuntu 系统中，直接执行: apt install libgl1